

Mohamed Mahmoud

Data Engineer

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SUMMARY

Computer science student specializing in data engineering, with hands-on experience building ETL pipelines, real-time alert systems, and cloud-based data architectures. Currently interning at the Digital Egypt Pioneers Initiative (DEPI), working with Apache Spark and AWS. Shipped three progressive versions of a production-grade data pipeline — from a naive Python script to an Airflow-orchestrated system to a real-time Redis-backed alert layer. Strong programming background from 100+ shipped Discord bots and a self-built AI bot generation platform.

EDUCATION

Alexandria University, Faculty of Science	2023 – 2027 (Expected)
<i>B.Sc. in Software Industry and Multimedia (SIM)</i>	<i>Alexandria, Egypt</i>
Victoria College	2020 – 2023
<i>High School Diploma</i>	<i>Alexandria, Egypt</i>

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, C#, Node.js
Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, ETL/ELT design, Data Modeling
Databases: PostgreSQL, MongoDB, Cassandra, Elasticsearch
Cloud & Infrastructure: AWS (S3, EC2), Docker
Tools: Git, GitHub, Linux, REST APIs, BeautifulSoup, Redis
Concepts: Distributed systems, OOP, Data Structures, Schema Design

EXPERIENCE

Digital Egypt Pioneers Initiative (DEPI)	2025 – Present
<i>Data Engineer Intern</i>	<i>Egypt</i>
- Building ETL pipelines with Apache Spark to process and transform structured datasets - Designing data models and optimizing query performance for analytical workloads on AWS - Collaborating with senior engineers to translate business requirements into pipeline specifications and reusable data assets	
Teleperformance	Jul 2025 – Oct 2025
<i>Customer Service Representative</i>	<i>Egypt</i>
- Resolved customer issues across CRM platforms while meeting daily case-volume and quality targets - Identified recurring issue patterns from case data and surfaced them to team leads to improve resolution scripts - Developed strong written and verbal communication in a fast-paced operational environment	
Target for Marketing and Collection	2023 – 2024
<i>Data Entry Supervisor</i>	<i>Egypt</i>
- Supervised data entry team handling large client datasets, enforcing accuracy and consistency standards - Implemented validation checks and review procedures that reduced downstream data correction work - Trained new hires on data entry workflows and quality control practices	
Self-Employed	2022 – 2023
<i>Freelance Discord Bot Developer</i>	<i>Remote</i>
- Designed and shipped 100+ custom Discord bots covering moderation, music, automation, and utility use cases - Built RESTful APIs in Node.js / Express with MongoDB for persistent bot state and user data - Integrated third-party payment gateways and AI APIs to extend bot functionality for paying clients	

PROJECTS

Real-Time Streaming Data Pipeline | *Kafka, Spark Streaming, Cassandra, Elasticsearch*

- Building an end-to-end streaming pipeline that ingests events through Kafka and processes them with Spark Streaming
- Writing processed events to Cassandra for low-latency reads and indexing to Elasticsearch for search and analytics
- Designing the system around fault tolerance, schema evolution, and horizontal scalability of each component

Weather Data Pipeline | *Python, Apache Airflow, PostgreSQL* | [GitHub](#)

DEPI Capstone

- Building an automated ETL pipeline that extracts weather data from a public API on a scheduled DAG, as the capstone project for the DEPI Data Engineering track
- Orchestrating extract, transform, and load stages in Airflow with retries, logging, and failure alerting
- Modeling the target schema in PostgreSQL to support time-series queries and historical comparisons

Egypt Jobs Pipeline | *Python, Airflow, PostgreSQL, Redis, Docker, Streamlit* | [GitHub](#)

V3 Complete

- Built a three-version data pipeline scraping Egyptian tech job listings from Wuzzuf — V1 as a naive Python script, V2 as an Apache Airflow DAG with 2-hour scheduling, exponential backoff retries, Discord webhook alerting, and a Streamlit dashboard
- Designed URL-based deduplication eliminating redundant Postgres inserts across scheduling cycles; containerized the full Airflow stack with Docker Compose and implemented backfill support for historical intervals
- Built V3 as a real-time alert layer — crash-proof polling loop checking Wuzzuf every 3 minutes, Redis SET deduplication for sub-millisecond seen-job lookups, keyword filters by role and seniority, and instant Discord notifications on new listings

Botify – AI-Powered Discord Bot Generator | *Node.js, MongoDB, AI APIs* | [GitHub](#)

- Built a platform that auto-generates custom Discord bots from client specifications using AI
- Implemented payment processing, encrypted storage of client configuration data, and a MongoDB-backed user system
- Productized a freelance workflow into a self-serve tool

TAWSEEL – University Transportation Web App | *HTML, CSS, JavaScript*

- Built a campus transportation web app for Alexandria University students with interactive route maps and live schedules
- Designed a responsive UI tested across desktop and mobile breakpoints

Five Nights at Faculty of Science | *Unity, C#*

- Collaborated on a horror-puzzle game inspired by FNAF, applying OOP patterns and Git-based version control
- Implemented dynamic lighting and audio systems for atmospheric gameplay

CERTIFICATES

Associate Data Engineer Certification – DataCamp, 2026

Associate Data Engineer in SQL – DataCamp (28-hour career track), 2026

Database Fundamentals – Information Technology Institute (ITI), 2025

Python Programming for AI – Alexandria University, 2024

VOLUNTEER WORK

Volunteer – Maaedat Farha